Fractional Knapsack - Greedy

Object -	1	2	3	4	5	6	7
Profit -	5	10	15	7	8	9	4
Weight -	1	3	5	4	1	3	2

Total Capacity = 15

$n = 7$

Method-1 - Max Profit

Object	Profit	Weight	Remaing Weight
3	15	5	15 − 5 = 10 (Total wt)
2	10	3	10 − 3 = 7
6	9	3	7 − 3 = 4
5	8	1	4 − 1 = 3
4	$7 \times \frac{3}{4} = 5.25$	3	3 − 3 = 0

← Start from Max weight

← obj 4 has weight = 4 but we only had Capacity = 3 left. so we will take $\left[\frac{3}{4}\right]$ of the profit/weight for obj = 4

$\boxed{\text{Total Profit} = 47.25}$

Method-2 Min Weight

Object	Profit	Weight	Remaing Weight
1	5	1	15 − 1 = 14
5	8	1	14 − 1 = 13
7	4	2	13 − 2 = 11
2	10	3	11 − 3 = 8
6	9	3	8 − 3 = 5
4	7	4	5 − 4 = 1
3	$15 \times \frac{1}{5} = 3$ (weight / weight)	1	1 − 1 = 0

$\boxed{\text{Total Profit} = 46}$

Method-3 Max $\frac{\text{Profit}}{\text{Weight}}$ ratio

Object	1	2	3	4	5	6	7
Profit	5	10	15	7	8	9	4
Weight	1	3	5	4	1	3	2
P/w	5	3.3	3	1.75	8	3	2

Object	Profit	Weight	Remaing Weight
5	8	1	15-1=14
1	5	1	14-1=13
2	10	3	13-3=10
3	15	5	10-5=5
6	9	3	5-3=2
7	4	2	2-2=0
	Total Profit = 51		

$\boxed{\frac{P}{W} > \text{Max Profit} > \text{Min Weight}}$

Use Max $\frac{P}{W}$ ratio